

Affective visualization graphs

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Setting up

```
# read in packages
library(tidyverse) # general use
library(here) # file organization -- dont need unless sort into folders
library(janitor) # cleaning data frames
```

```
library(DHARMA) # for diagnostics
library(ggeffects) # for ggpredict
library(flextable) # for flextable
library(gtsummary)
```

```
# read in personal data to object from csv
my_data <- read_csv("envs193ds-personal-data.csv")
```

Framing my question

My central question is: How does sleep duration the night prior influence the probability of me going on a run the next day?

Variables

- Predictor: sleep duration the previous night (hours)
- Response: probability of going on a run (1/0) (1 = yes; 0 = no run)

Cleaning and wrangling data

```
# cleaned column names
my_data <- my_data |>
  clean_names()

# created new column: 1 if workout was a run, 0 if not
my_data <- my_data |>
  mutate(is_run = case_when(
    workout_type == "Run" ~ 1, # run = 1
    .default = 0))             # all other workout types = 0
```

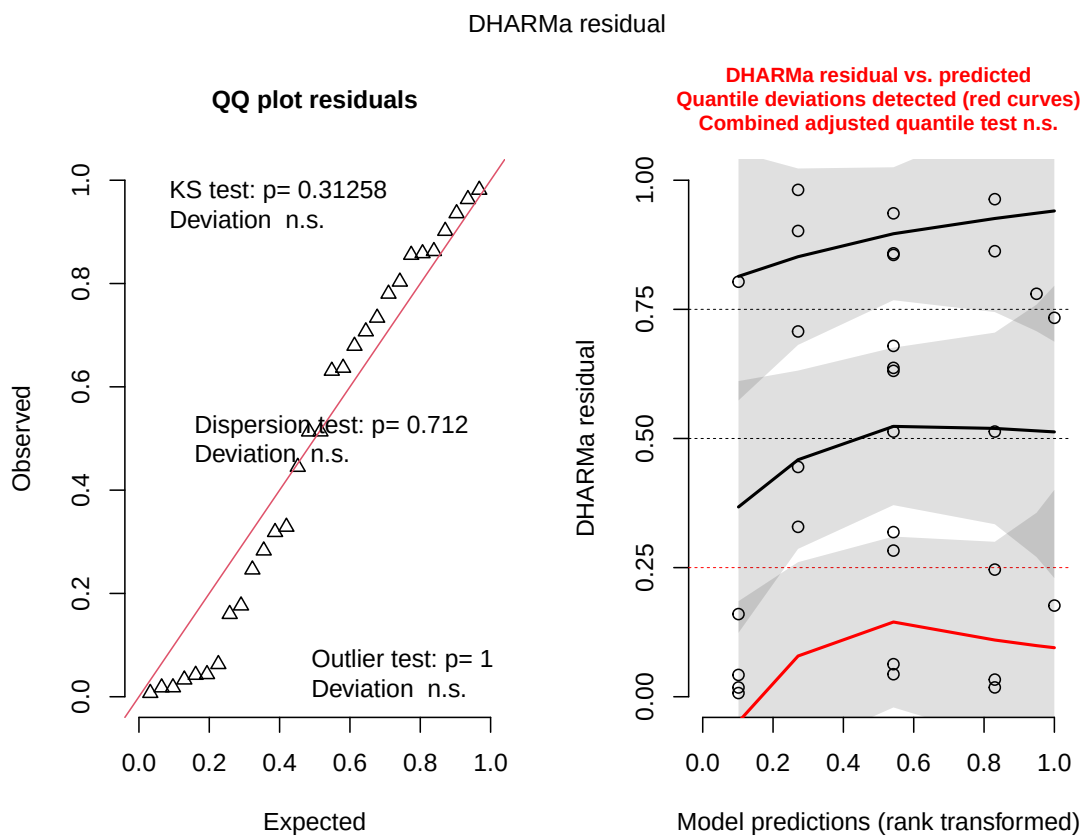
Fitting the model

```
# fitted generalized linear model for running
# response: whether or not workout was a run (binary: 1/0)
# predictor: sleep the previous night (hours)
run_mod <- glm(
```

```
# formula: binary run outcome ~ sleep hours
is_run ~ sleep_hours,
# data: my_data data frame
data = my_data,
# binomial error distribution for binary response
family = binomial)
```

Checking diagnostics

```
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(run_mod))
```



Info from left plot (QQ plot of residuals):

- KS test: do my residuals follow a uniform distribution? → generally yes (CHECK)

- Dispersion test: are your residuals more dispersed than expected from a uniform distribution? → no
- Outlier test: are there outliers in your data set that could be influencing model coefficients or predictions? → no

Info from right plot (residuals vs predicted values):

- By default, R generates the reference lines for quartiles: 25, 50, 75
- Not that informative but look at patterns within residuals across the range → DHARMA diagnostics flag issues in 25th quartile due to small sample size, not because the model is wrong.

Looking at the model coefficients

```
# displayed model summary
summary(run_mod)
```

Call:

```
glm(formula = is_run ~ sleep_hours, family = binomial, data = my_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.7011	3.4964	-2.489	0.0128 *
sleep_hours	1.0873	0.4712	2.307	0.0210 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.652 on 29 degrees of freedom
 Residual deviance: 28.046 on 28 degrees of freedom
 AIC: 32.046

Number of Fisher Scoring iterations: 5

Calculate inverse logit

Characteristic	OR	95% CI	p-value
sleep_hours	2.97	1.37, 9.40	0.021

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
# annotate
plogis(-8.7011)
```

```
[1] 0.000166375
```

Note: - inverse logit is 0.00016 - The probability of going on a run if sleep duration is 0 hours is 0.00016??

Calculate odds ratio

```
# annotate
gtsummary::tbl_regression(run_mod, exponentiate = TRUE)
```

Note: - odds ratio is 2.97 - $2.97 > 1$ so the probability of going on a run increases by a factor of 2.97 with every 1 hour increase in sleep

Summary: - For every 1 hour increase in sleep, the the odds of going on a run increase by a factor of 2.97 (95% CI: [1.37, 9.40], $p = 0.021$, $\alpha = 0.05$). ?? NOT SIGNIF THO

Other way to calculate – remove?

```
# exponentiated coefficients to get odds ratios
# e.g. exp(coef) = odds ratio for 1 hr increase in sleep
exp(coef(run_mod))
```

```
(Intercept)  sleep_hours
0.0001663954 2.9663093920
```

Looking at the model predictions

- using probabilities to determine chance of getting certain outcome
- line represents the probability of 1 (going on a run) actually occurring

```
# what is the probability of going on a run after sleeping for 10 hours?  
ggpredict(run_mod,  
          terms = "sleep_hours [10]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
10	0.90	0.39, 0.99

```
# what is the probability of going on a run after sleeping for 4 hours?  
ggpredict(run_mod,  
          terms = "sleep_hours [4]")
```

Predicted probabilities of is_run

sleep_hours	Predicted	95% CI
4	0.01	0.00, 0.25

Note:

- If I slept for 10 hours, the probability of going on a run the next day is 0.90 (95% CI [0.39, 0.99]).
- If I slept for 4 hours, the probability of going on a run the next day is 0.01 (95% CI [0.00, 0.25]).

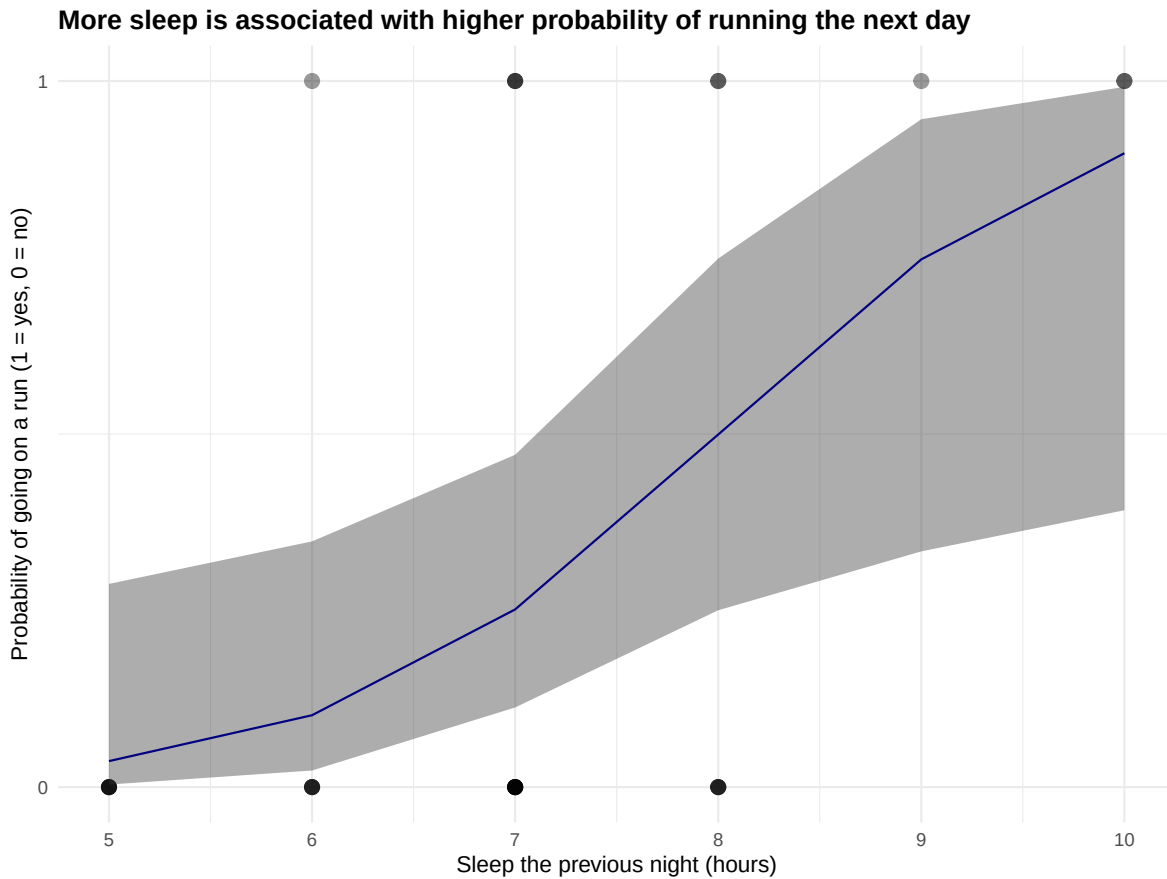
Visualizing model predictions

```
# generated model predictions across range of sleep values in data set  
mod_preds <- ggpredict(run_mod,  
                      terms = "sleep_hours [5:10 by = 1]")
```

```

# base layer: ggplot with raw data
# data frame: my_data
ggplot(my_data,
       aes(x = sleep_hours,
           y = is_run)) +
# first layer: raw observations
geom_point(size = 3,
           alpha = 0.4) +
# second layer: confidence interval ribbon from model predictions
geom_ribbon(data = mod_preds,
           aes(x = x,
               y = predicted,
               ymin = conf.low,
               ymax = conf.high),
           alpha = 0.4) +
# third layer: logistic curve from model predictions
geom_line(data = mod_preds,
          aes(x = x,
              y = predicted),
          # custom color
          color = "navyblue") +
# constrained y-axis to 0 and 1
scale_y_continuous(limits = c(0, 1),
                   breaks = c(0, 1)) +
# relabeled axes and added title
labs(title = "More sleep is associated with higher probability of running the next day",
     x     = "Sleep the previous night (hours)",
     y     = "Probability of going on a run (1 = yes, 0 = no)") +
# changed theme from default
theme_minimal() +
# bolded title
theme(plot.title = element_text(face = "bold"))

```



Describing the model

```
# displaying model in table
as_flextable(run_mod)
```

	Estimate	Standard Error	z value	Pr(> z)	
(Intercept)	-8.701	3.496	-2.489	0.0128	*
sleep_hours	1.087	0.471	2.307	0.0210	*

Estimate	Standard Error	z value	Pr(> z)
<i>Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '.' < 0.05</i>			

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 36.65 on 29 degrees of freedom

Residual deviance: 28.05 on 28 degrees of freedom